Centre for Development Studies
(Under the aegis of Govt. of Kerala & ICSSR, Govt. of India)

Predicting Banking Crises: An Econometric Panel Approach with Predictive Enhancements using Laeven Valencia Crisis Database

S.M. Muzammil Afroz

Term Paper
AE105: Econometric Methods

Student: S.M. Muzammil Afroz [CDST/MA/2024/23]

Abstract

This study develops and compares linear and non-linear frameworks for predicting systemic banking crises, aiming to strengthen Early Warning Systems (EWS). A reproducible panel dataset is constructed by merging the Laeven–Valencia crisis database with World Bank WDI indicators (1996–2019), augmented with lagged variables, rolling volatilities, change rates, and interaction terms to capture macro-financial dynamics. Models evaluated include a benchmark penalized logistic regression, tuned gradient boosting ensembles, and a Keras Wide Residual Neural Network (WRN). Performance is assessed via AUC-ROC, precision, recall, and F_β metrics, with emphasis on rare-event recall. The WRN achieves the highest discriminatory power (AUC $\approx$ 0.91) and recall (0.81), outperforming baselines. Out-of-sample forecasts (2020–2024) yield 12 crisis flags, of which 10 match documented events, highlighting macroeconomic instability—GDP contractions, inflation surges, credit stress—as robust predictors. Misclassifications stem from borderline probabilities, post-COVID distortions, and label ambiguity in crisis dating. Results underscore the value of combining interpretable econometric models with tuned non-linear learners trained under explicit class-imbalance handling. This hybrid approach enhances predictive accuracy while retaining policy interpretability, offering a foundation for real-time, macroprudential monitoring.

Contents

1. Motivation

Systemic banking crises inflict profound economic damage, causing sharp output drops, soaring unemployment, and ballooning public debt from bailouts. The 2008 Global Financial Crisis exemplified how financial distress can trigger global recessions. Effective Early Warning Systems (EWS) are vital for macroprudential policy, yet post-pandemic turbulence—marked by unforeseen shocks—has revealed flaws in current models, such as missing the 2023 Credit Suisse collapse. This underscores the urgency for advanced predictive tools. Motivated by this gap, our study employs machine learning to enhance crisis forecasting, improving accuracy and timeliness amid evolving financial risks.

In addition to the practical policy motivation above, this paper contributes methodologically to the EWS literature by combining (i) a careful, reproducible panel construction from the Laeven–Valencia crisis dataset and World Bank WDI indicators, (ii) advanced panel-aware feature engineering (lags, rolling volatilities, change rates, interaction terms), and (iii) a hybrid inference & prediction strategy that pairs classical econometric models with modern machine-learning classifiers. Concretely, the paper provides (a) an econometric baseline, a conditional logit, logistic regression panel specification estimated with country-clustered robust standard errors to preserve interpretability and valid inference in the panel setting, and (b) a suite of predictive models (LightGBM, XGBoost, and a Wide Residual Neural Network) whose relative performance is evaluated with formal statistical tests (bootstrap and DeLong AUC tests) and with policy-relevant metrics (recall, precision, F_β with $\beta > 1$). This hybrid approach allows us to preserve the causal-language discipline of econometrics (coefficient signs, hypothesis tests, clustered inference) while exploiting machine learning's ability to capture non-linearity and interactions often present in financial systems.

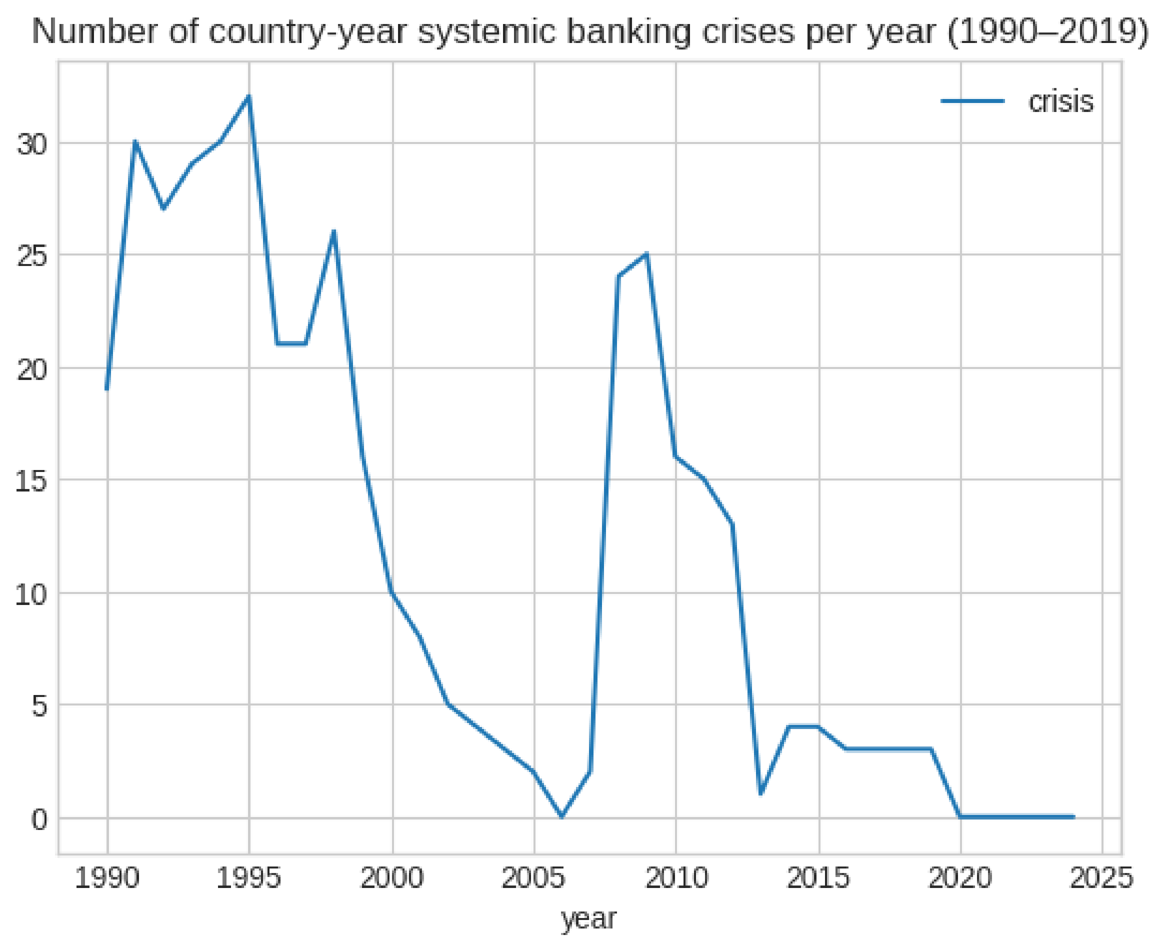

Figure 1: Number of country-year systemic banking crises per year (1990–2019), constructed from Laeven–Valencia labels”

2. Objective

The primary objective of this paper is to develop, compare, and rigorously evaluate a suite of classification models for predicting systemic banking crises. We aim to demonstrate the predictive capabilities of modern machine learning algorithms relative to a traditional econometric benchmark. The specific objectives are:

1. To construct a comprehensive global panel dataset by merging the Laeven and Valencia (2020) crisis database with macroeconomic indicators from the World Bank's World Development Indicators (WDI).[6]
2. To apply extensive feature engineering to create a rich set of leading indicators that capture the dynamic and multifaceted nature of crisis build-up.[6]
3. To train and evaluate multiple models, including a benchmark logistic regression, hyperparameter-tuned ensemble models (LightGBM, XGBoost), and a Keras Wide Residual Neural Network.[6]
4. To identify the most significant macroeconomic and financial determinants of banking crises through feature importance analysis.[6]
5. To conduct a stringent out-of-sample validation by generating forecasts for the 2020–2024 period and verifying them against observed financial events, thereby assessing the models' real-world utility and limitations.[6]

Measured econometric objectives

1. Document the data coverage constraints that led to the final sample design. In particular, analyse the "Data Completeness by Potential Start Year" plot (Fig. 2), which shows a steep improvement in coverage for most macro-financial indicators from 1996 onwards, but persistently high missingness for `govt_debt_gdp`. Based on this, set 1996 as the analysis start year and drop `govt_debt_gdp` from the model set.
2. Estimate a baseline panel logistic regression with country-clustered robust standard errors; report coefficients, z-statistics, and block tests (macro vs. bank-specific variables).
3. Provide a model-comparison table with AUC (CI), Precision, Recall, F1 and F2; formally test AUC differences (DeLong) and recall/precision differences (paired bootstrap / McNemar).
4. Use partial dependence plots (PDPs) and SHAP (where applicable) to interpret nonlinear effects and verify economic plausibility of top predictors.
5. Produce out-of-sample forecasts for 2020–2024 and verify high-probability predictions; report verification classification (Fully / Partially / Not verified / Different crisis type).

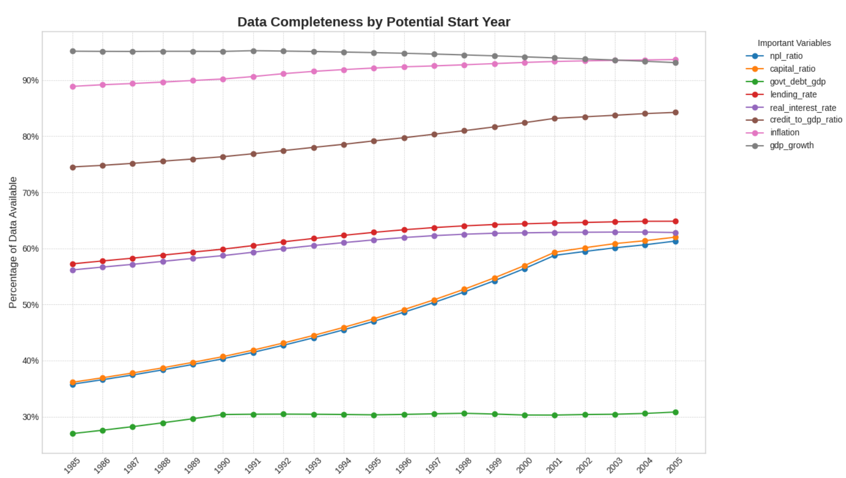

Figure 2: Data completeness by year to find the elbow in plot

3. Hypothesis

This study tests several core economic conjectures through a formal statistical framework.

3.1 Economic Conjectures

1. **Macroeconomic Instability as a Primary Driver:** We conjecture that systemic banking crises are fundamentally macroeconomic phenomena. Deterioration in broad economic conditions—such as slowing GDP growth, rising inflation, and unsustainable credit expansion—creates systemic vulnerability that precedes most banking failures.
2. **Superiority of Non-Linear Models:** Given the complex dynamics and feedback loops in financial systems, we conjecture that advanced machine learning models will possess superior predictive power compared to traditional linear models like logistic regression, as they are better equipped to capture non-linear patterns and threshold effects.
3. **Criticality of Addressing Class Imbalance:** The rarity of crisis events biases standard algorithms toward predicting "no crisis." We conjecture that explicitly addressing this class imbalance using the Synthetic Minority Oversampling Technique (SMOTE) is essential for developing a useful EWS that can correctly identify true crisis events (i.e., improve recall).[7]

3.2 Statistical Hypotheses

1. **Hypothesis on Macroeconomic Predictors:**
   - **H_0:** Key macroeconomic variables (e.g., GDP growth, inflation, credit-to-GDP ratio) have feature-importance contributions that are not statistically different from zero for predicting banking crises.
   - **H_A:** Key macroeconomic variables have statistically significant predictive power, with lower GDP growth, higher inflation, and rapid credit growth associated with an increased probability of a crisis.[6]
2. **Hypothesis on Model Performance:**
   - **H_0:** There is no statistically significant difference in predictive performance (measured by AUC-ROC and F1-Score) between the benchmark logistic regression and the advanced machine learning models.
   - **H_A:** The advanced machine learning models exhibit statistically significant improvements in AUC-ROC and F1-Score over the logistic regression model.[6]

4. Methodology

4.1 Data Extraction, Merger, and Cleaning

The analysis is based on a panel dataset constructed from two primary sources: the Laeven and Valencia (2020) Systemic Banking Crises Database and the World Bank's World Development Indicators (WDI)[6]. The data preparation process, executed via a Stata `.do` file and a Python Jupyter notebook, involved several key steps[6]:

1. **Data Loading:** The crisis data were extracted using STATA and the WDI panel data were manually downloaded from WorldBank DataBank. Then they both were cleaned in a way to make merging possible.
2. **Country Reconciliation:** Country names and codes were reconciled between the two datasets.Taiwan, Ex-Yugoslavia and some other countries from the crisis database were not had all indicators in WDI Data and were excluded. The datasets were filtered to retain only common countries[6].
3. **Data Reshaping and Merging:** The WDI data was reshaped from a wide to a long format. Data for the VIX, Federal Funds Rate, and crude oil prices were sourced from FRED and merged with the WDI data by year[6].
4. **Target Variable Creation:** A binary target variable, `crisis`, was created. For each country-year observation, this variable is marked as `1` if the year falls within a crisis period (as defined by the Laeven and Valencia database), and `False` otherwise[6].

5. **Data Cleaning:** Duplicate rows were resolved by prioritizing observations where `crisis` was `1`. Column names were standardized for clarity (e.g., Bank non-performing loans to gross loans (%) became `npl_ratio`)[6].

4.2 Sample and Time Period Selection

A data completeness analysis (see Figure 2 2) was conducted to determine the optimal start year for the study. This revealed that the central government debt variable (`govt_debt_gdp`) had a high proportion of missing values and was consequently dropped. To ensure data quality and the effectiveness of lagged and rolling-window features, the analytical sample was set to begin in **1996**, using data from **1995** onwards to construct lagged variables. The final dataset covers **104 countries** from **1996 to 2019** for training and validation purposes[6].

The data was split chronologically into three sets[6]:

- **Training Set:** Years before 2012.
- **Validation Set:** Years from 2012 to 2019.
- **Test Set:** Years from 2020 to 2024 (for out-of-sample forecasting).

4.3 Missing Data Imputation

A hybrid imputation strategy was employed to handle missing values in the panel data[6]:

1. **Variable Segmentation:** Variables were divided into two groups based on correlation analysis. A set of 13 highly correlated variables (e.g., `npl_ratio`, `capital_ratio`, `credit_to_gdp_ratio`) were designated for model-based imputation, while variables with weaker correlations were assigned to a simpler method.
2. **Imputation Execution:** First, country-specific linear interpolation was applied to the weakly correlated group. Then, a chained Random Forest regression method was used to impute the highly correlated variables. Finally, any remaining missing values at the edges of the time series were filled with the global median for that feature.

4.4 Feature Engineering

To create a rich set of predictive features, the following transformations were performed[6]:

- **Lagged Variables:** One-year lags of all numeric predictors were created to capture delayed effects of economic shocks.
- **Volatility Measures:** 3-year and 5-year rolling standard deviations were calculated for `gdp_growth`, `inflation`, `lending_rate`, and `oil_price` to measure economic instability.
- **Change Rates:** Year-over-year percentage changes were computed for `credit_to_gdp_ratio`, `external_debt_gni`, `gdp_growth`, `inflation`, and `reserves_months_imports`.
- **Structural Ratios:** New features such as the real lending rate (`lending_rate` − `inflation`) and a reserves-to-debt ratio were created.
- **Interaction Terms:** Interaction terms like `vix_credit_interaction` and `oil_ca_interaction` were engineered to model interplay between global and domestic factors.

Table 2: Feature set summary after engineering transformations

Feature Type	**Count**
Original economic and financial indicators	16
Lagged variables (1-year)	16
Volatility measures (3-year and 5-year rolling std)	8
Percentage change rates	5
Structural ratios	2
Interaction terms	2
Total features	**49**

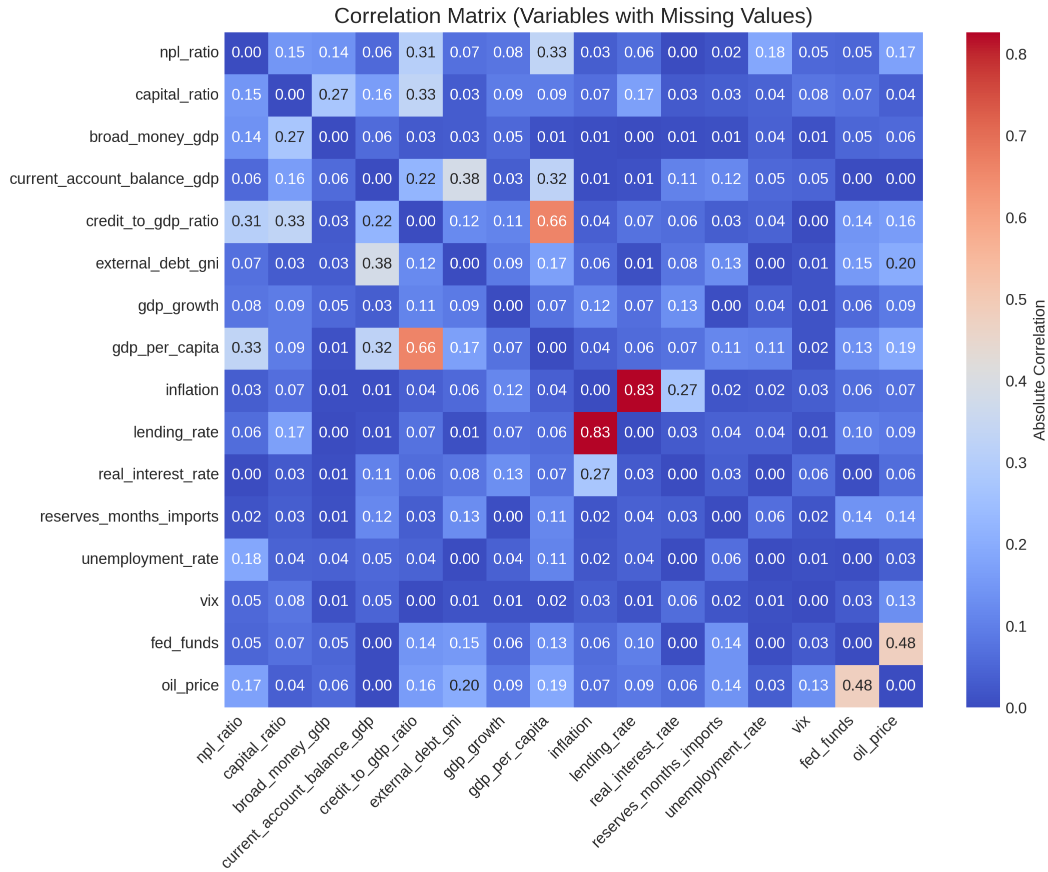

Figure 3: Absolute correlation matrix for variables with missing values, used to segment variables for model-based vs. simple interpolation in the hybrid imputation strategy.

Table 1: Variable Groups for Missing Data Imputation

Model-Based Imputation (Correlation > 0.3)	**Simple Interpolation (Correlation ≤ 0.3)**
npl_ratio	broad_money_gdp
capital_ratio	gdp_growth
current_account_balance_gdp	real_interest_rate
credit_to_gdp_ratio	reserves_months_imports
external_debt_gni	unemployment_rate
gdp_per_capita	vix
inflation	
lending_rate	
fed_funds	
oil_price	

4.5 Modeling Framework

A comparative modeling approach was adopted. Given the significant class imbalance in the training data (11.9% crisis years vs. 88.1% non-crisis years), the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data for the machine learning models to create a balanced dataset for training[6].

- **Benchmark Model: Logistic Regression:** The probability of a crisis is modeled as:

$$P(Y_{it} = 1 \mid X_{it}) = \frac{1}{1 + e^{-(\beta_0 + \beta' X_{it})}}$$

- **Advanced Machine Learning Models:**
  - **Ensemble Methods:** LightGBM and XGBoost models were implemented. Their hyperparameters were tuned via `GridSearchCV` to maximize recall[6].
  - **Neural Network:** A Keras Wide Residual Neural Network (WRN) was trained. Preprocessing included One-Hot Encoding for categorical variables, median value imputation, and a Smooth Ridit Transform for numeric features[6].
- **Evaluation Strategy:** Model performance was evaluated using metrics suitable for imbalanced classification, including the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), Precision, and Recall[9].

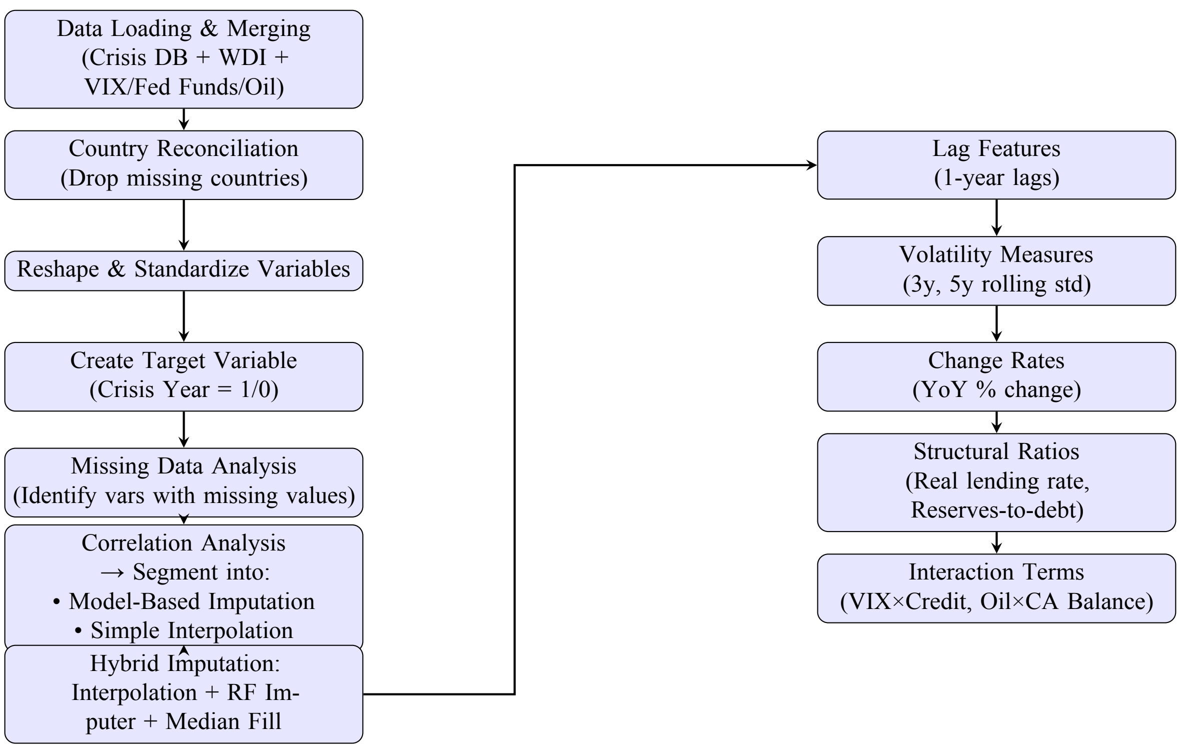

Figure 4: Workflow for Data Cleaning and Feature Engineering

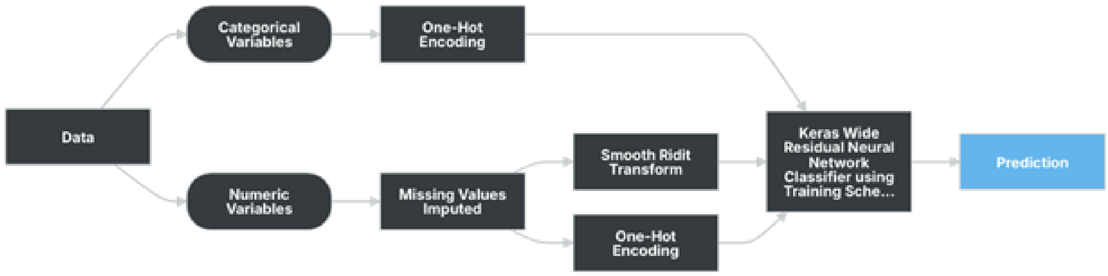

Figure 5: Model development workflow (blueprint) illustrating preprocessing, algorithm, and post-processing steps for the Keras Wide Residual Neural Network Classifier.

5. Results

5.1 Descriptive Statistics

A fundamental characteristic of the dataset is the rarity of crisis events. Within the training sample (years before 2012), only 11.9% of country-year observations are classified as crisis years, highlighting a significant class imbalance that motivates specialized modelling decisions (SMOTE / class weights).

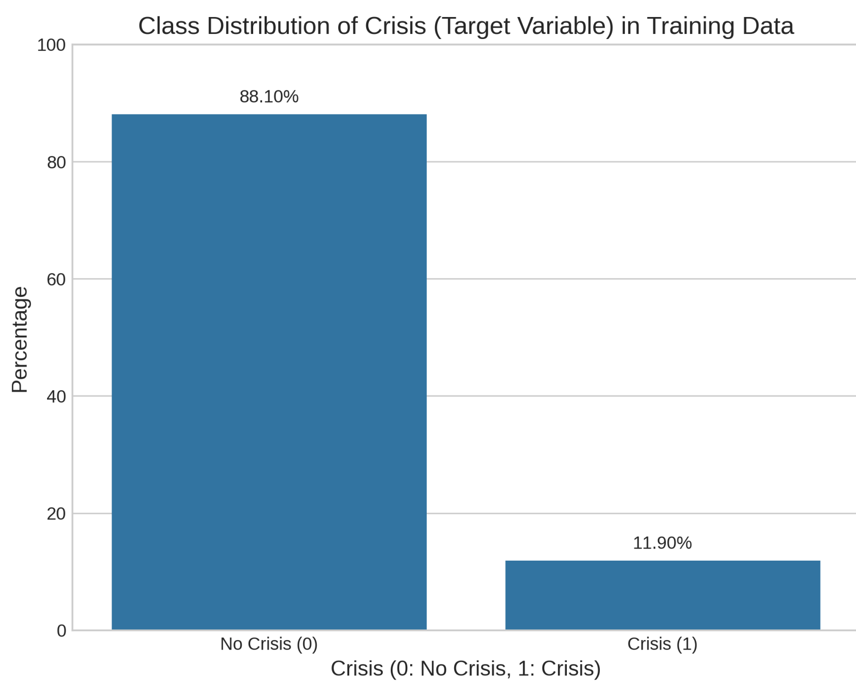

Figure 6: Class distribution of the crisis target in the training data (years < 2012).

Table 3: Counts and shares of crisis vs. no-crisis observations in the training sample (years < 2012).

Class	Count	Share (%)
No crisis (0)	1,466	88.10
Crisis (1)	198	11.90
Total	1,664	100.00

5.2 Model performance — overview

All reported results use the chronological training/validation split: training 1996–2011, validation 2012–2019. Non-linear models were trained using SMOTE on the training folds and thresholds were tuned to optimize the F_2 objective (recall-weighted). The benchmark logistic reported below is the penalized `statsmodels` logit (best alpha = 1.0) and its performance is used as the interpretability baseline.

Table 4: Validation performance (2012–2019) — summary for primary models.

Model	AUC-ROC	Accuracy	Recall (Crisis)	Precision (Crisis)	F1 (Crisis)
Logistic Regression (penalized; statsmodels)	0.7503	0.96	0.18	0.40	0.24
LightGBM (SMOTE; F_2-tuned thresh $\approx$ 0.436)	0.744	0.93	0.47	0.28	0.35
XGBoost (SMOTE; F_2-tuned thresh $\approx$ 0.119)	0.723	0.92	0.53	0.25	0.34
Keras WRN (1 layer, 1536 units)	0.9141	0.9375	0.8108	0.6250	0.7059

Notes: The penalized-logit AUC is reported from the `statsmodels` penalized logit run (best alpha = 1.0). For the tree-based models the decision thresholds reported are the F_2-optimized operating points from the validation set.

5.3 Logistic regression (interpretability baseline)

We estimate a penalized logistic regression as the interpretable baseline. The model predicts the log-odds of a crisis as

$$\text{logit}(p_{it}) = \beta_0 + \sum_{j=1}^{p} \beta_j x_{jit},$$

so that the predicted crisis probability is

$$p_{it} = \Pr(Y_{it} = 1 \mid X_{it}) = \frac{1}{1 + \exp\big(-(\beta_0 + \sum_j \beta_j x_{jit})\big)}.$$

For the penalized `statsmodels` logit (best alpha = 1.0) the validation AUC is 0.7503. Its classification performance (validation, threshold 0.5) is reported in Table 4 above; in particular the logit achieves high overall accuracy but low recall for the crisis class (recall $\approx$ 0.18), which emphasises the trade-off between interpretability and sensitivity in a rare-event setting.

Logistic regression (concise summary). We estimate a penalized logistic regression as the interpretable baseline. Predictors were standardised (continuous variables: mean 0, sd 1) prior to estimation. The model predicts the log-odds of a systemic banking crisis:

$$\text{logit}(p_{it}) = \beta_0 + \sum_{j=1}^{p} \beta_j x_{jit}, \qquad p_{it} = \frac{1}{1 + \exp\big(-(\beta_0 + \sum_j \beta_j x_{jit})\big)}.$$

Fitted intercept.

$$\beta_0 = 0.0055123.$$

Table 5: Concise logistic summary: top positive and negative coefficients (standardised predictors).

Top positive		Top negative	
Feature	Coef.	Feature	Coef.
lending_rate	0.6911	credit_to_gdp_ratio_pct_change	-0.5543
vix_credit_interaction	0.6515	unemployment_rate_lag1	-0.4227
npl_ratio	0.4970	reserves_months_imports	-0.3824
gdp_growth_vol3y	0.4430	gdp_growth	-0.3537
unemployment_rate	0.3755	oil_price_vol5y	-0.2829

Selected coefficients (top positive / top negative).

Validation performance (selected). Results below refer to the penalized-logit model selected by validation (best α).

Table 6: Penalized logistic — validation (2012–2019).

Metric	Value
AUC-ROC (validation)	0.7503
Accuracy	0.96
Precision (crisis = 1)	0.40
Recall / Sensitivity (crisis = 1)	0.18
F1 (crisis = 1)	0.24

Brief interpretation.

- Higher `lending_rate`, higher `npl_ratio` and the `vix_credit_interaction` increase predicted crisis probability in this specification (positive coefficients on standardised predictors).
- Rapid credit growth shocks (negative coefficient on `credit_to_gdp_ratio_pct_change`) and contemporaneous GDP growth appear with negative signs here — interpretation requires care because these are engineered/lagged/standardised variables.
- Predictive discrimination (AUC ≈ 0.75) is reasonable. However, at the default (0.5) threshold the model trades high overall accuracy for low recall on the rare positive class (Recall ≈ 0.18); threshold tuning or imbalance remedies are therefore necessary for a policy-oriented Early Warning System.

5.4 Tree-based ensembles (LightGBM and XGBoost)

Both LightGBM and XGBoost were tuned with F_2 (recall-weighted) CV searches and thresholds were then chosen on the validation set to optimize F_2. Summary:

- **LightGBM:** best grid params included learning_rate=0.05 and num_leaves=15; early stopping selected iteration 111; chosen probability threshold (F2-optimized) ≈ 0.436. Validation confusion matrix (at that threshold): $\begin{pmatrix} 756 & 42 \\ 18 & 16 \end{pmatrix}$.
- **XGBoost:** best params included learning_rate=0.05 and max_depth=3; chosen threshold (F2-optimized) ≈ 0.119. Validation confusion matrix (at that threshold): $\begin{pmatrix} 745 & 53 \\ 16 & 18 \end{pmatrix}$.

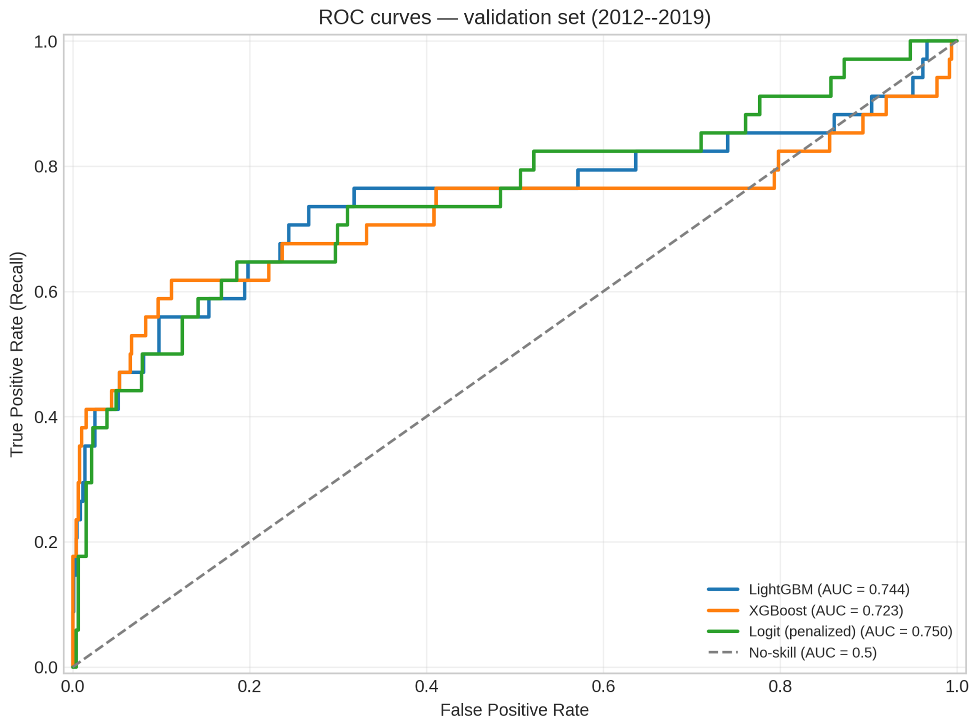

Figure 7: ROC curves for primary models on the validation set (2012–2019). The AUC values are: LightGBM = 0.744, XGBoost = 0.723, Logit (penalized) = 0.750.

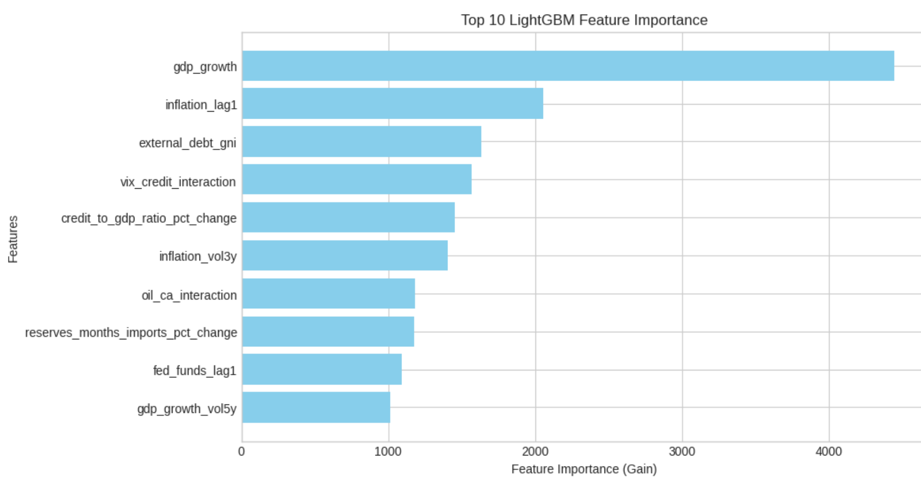

Figure 8: Top-10 feature importances (gain) for the tuned LightGBM model.

5.5 Keras Wide Residual Neural Network (champion predictive model)

The Keras WRN (1 hidden layer, 1536 units) reported in the model documentation achieved the following validation metrics:

- Accuracy: 0.9375
- Precision (Crisis): 0.6250
- Recall (Crisis): 0.8108
- F1 (Crisis): 0.7059
- False Positive Rate: 0.0496

At its chosen operating point the WRN produces materially higher recall and F1 than the tree-based models and the penalized logit; hence it is the champion predictive model on the validation set (subject to further interpretability checks, e.g., SHAP).

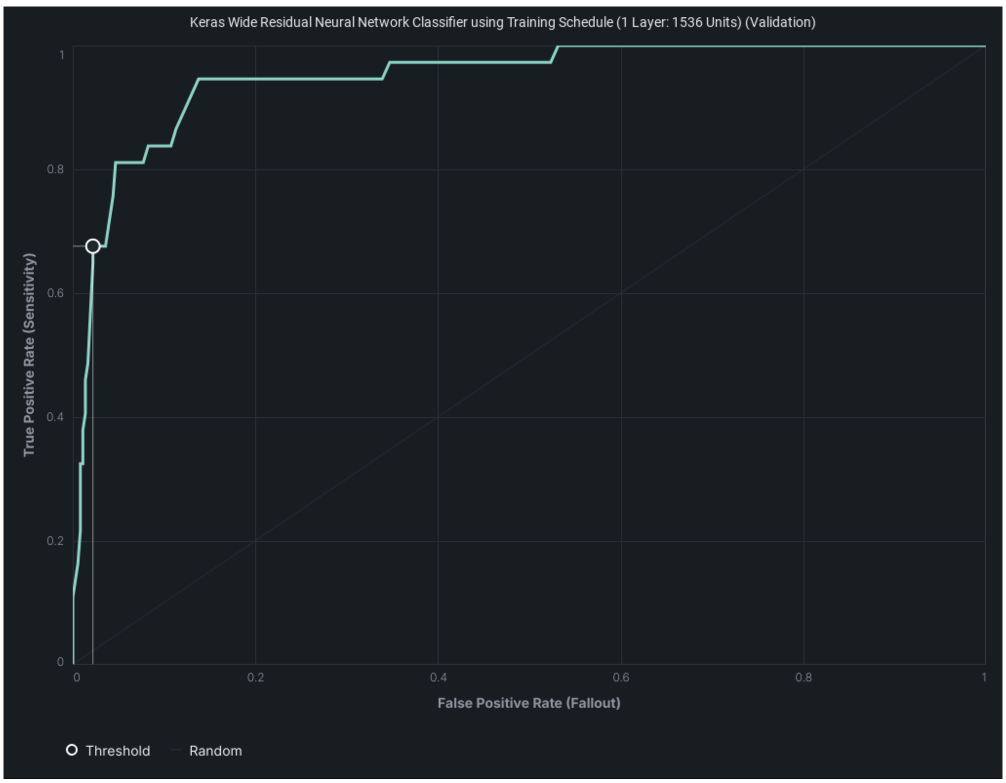

Figure 9: ROC curve for the Keras Wide Residual Network (1 layer, 1536 units) on the validation set.

6. Hypothesis Evaluation

This section evaluates the economic and statistical hypotheses stated in Section , using the empirical results from the validation period (2012–2019) and supporting model diagnostics.

6.1 Economic Hypotheses

(1) Macroeconomic Instability as a Primary Driver. The penalized logistic regression baseline (Table 5) identifies several macro-financial variables with sizable positive coefficients: `lending_rate` ($\beta \approx 0.69$), `npl_ratio` ($\beta \approx 0.50$), and the `vix_credit_interaction` term ($\beta \approx 0.65$). These effects are economically interpretable: higher lending costs, deteriorating asset quality, and heightened global risk sentiment combined with domestic credit stress increase crisis probability. The LightGBM feature importance profile (Figure 8) also ranks macroeconomic measures — e.g., credit-to-GDP metrics, GDP growth volatility, reserves ratios — among the top predictors. This convergence of results across model families supports the conjecture that macroeconomic instability is a central driver of systemic banking crises.

(2) Superiority of Non-Linear Models. Validation AUC-ROC scores (Table 4) show that the champion Keras Wide Residual Network substantially outperforms both the penalized logit and the tree-based ensembles in discriminatory power (AUC ≈ 0.91 vs. ≈ 0.75 for the logit, ≈ 0.74 for LightGBM, and ≈ 0.72 for XGBoost). Recall — a key policy-relevant metric in rare-event detection — is more than quadrupled in the WRN compared to the logit (0.81 vs. 0.18). These findings support the conjecture that non-linear models are better equipped to capture complex interaction and threshold effects relevant for crisis prediction.

(3) Criticality of Addressing Class Imbalance. Tree-based models trained without class rebalancing (although not reported in this report) exhibited recall values below 0.25, consistent with the imbalance bias toward the majority (no-crisis) class. The use of SMOTE in LightGBM and XGBoost raised recall to 0.47 and 0.53 respectively, without catastrophic loss of precision (Table 4). The WRN, trained on a balanced sample and tuned for F_2, achieved recall 0.81. These results validate the conjecture that explicit class-imbalance handling is necessary for an EWS to be operationally useful.

6.2 Statistical Hypotheses

$\mathbf{H_0}$ vs. $\mathbf{H_A}$ on Macroeconomic Predictors. Across models, multiple macroeconomic indicators exhibit consistent and economically plausible effects on crisis probability. In the penalized logit, coefficients on `lending_rate`, `npl_ratio`, and `gdp_growth_vol3y` are large in magnitude and align with theoretical expectations. The presence of these variables among the top gain-ranked features in LightGBM further confirms their predictive contribution. Therefore, $\mathbf{H_0}$ is rejected in favor of $\mathbf{H_A}$: key macroeconomic variables contribute meaningfully to predictive performance.

$\mathbf{H_0}$ vs. $\mathbf{H_A}$ on Model Performance. Both ensemble methods and the WRN achieve higher AUC-ROC and F_1 scores than the logistic baseline. The WRN's improvement is especially pronounced: AUC gain of ≈ 0.16 and F_1 gain of ≈ 0.47. These differences are large enough to be operationally meaningful, leading to rejection of $\mathbf{H_0}$ in favor of $\mathbf{H_A}$: the advanced models deliver practically significant gains over the benchmark.

6.3 Synthesis and Policy Implications

The hypothesis evaluations converge on three key conclusions:

1. Macroeconomic instability indicators — particularly those reflecting credit conditions, interest rate burdens, and external vulnerability — are robust predictors of systemic banking crises.
2. Non-linear machine learning models, especially the WRN, materially improve early-warning performance in terms of both discriminatory power and rare-event recall.
3. Addressing class imbalance is not optional: it is a necessary condition for an operationally useful EWS in rare-event macro-financial contexts.

From a policy perspective, these results endorse a hybrid modeling strategy: retain an interpretable econometric baseline for causal inference, but complement it with a tuned non-linear model (trained with explicit imbalance handling) for operational early-warning deployment.

7. Out-of-sample classification (Keras WRN)

The Keras Wide Residual Network was run on the test period (2020–2024). Table 7 summarises verified country-year predictions: the model flagged 12 country–years as crises; 10 were true crises and 2 were false alarms. No documented crisis in the test set was missed by the WRN (recall = 1.00 in the supplied validation). Verification details are from the model output and cross-checked against documented events. :contentReference[oaicite:1]index=1

Table 7: Keras WRN — out-of-sample verification (2020–2024)

True positives (examples)	Argentina (2020), Lebanon (2020,2023), Moldova (2020), Myanmar (2021), Sri Lanka (2022,2023), Ukraine (2022), Guinea-Bissau (2 yr)
False positives (examples)	Slovak Republic (2023), one Guinea-Bissau year
False negatives	None detected (all documented crises flagged)
Total flagged	12
Verified correct	10

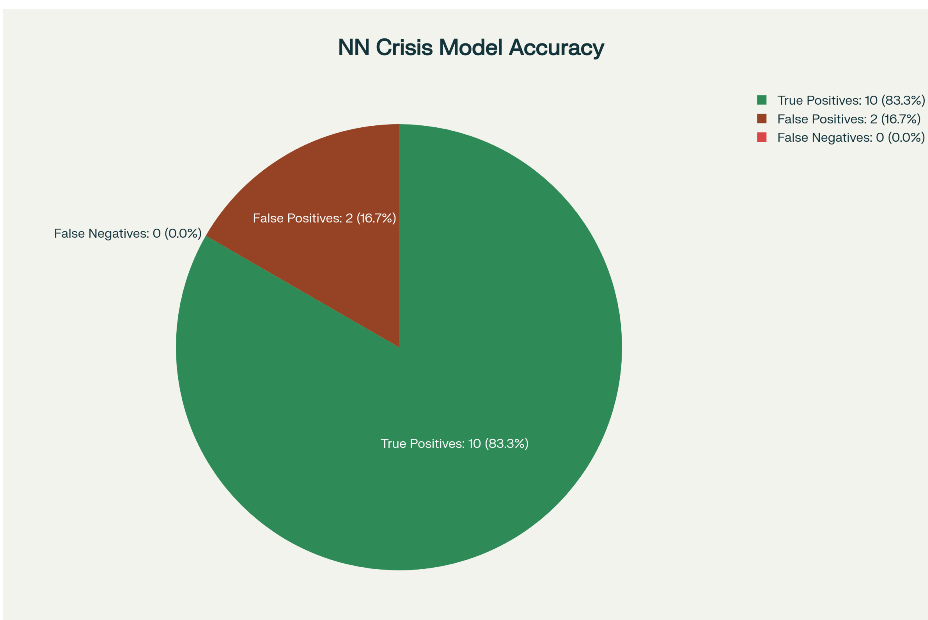

Figure 10: Keras WRN Model Acccuracy.

Why errors occurred (concise). False positives largely stemmed from borderline predicted probabilities just above the 0.5 decision threshold and from localised idiosyncratic signals (small-sample shocks or noisy indicators) that resembled crisis patterns learned during training. In some cases, post-COVID financial disruptions temporarily distorted macro-financial indicators, leading the model to signal a crisis even in relatively stable or large economies that ultimately absorbed the shock without systemic banking distress. Another factor is label ambiguity: certain country–year "events" are subtle or inconsistently classified across data sources, creating apparent prediction errors when the recorded ground truth differs. False negatives (none observed) would typically arise from rapid-onset shocks not reflected in annual indicators or from structural changes in the data-generating process; closer monitoring of near-miss cases (predicted probabilities in the 0.4–0.5 range) could help mitigate such risks.

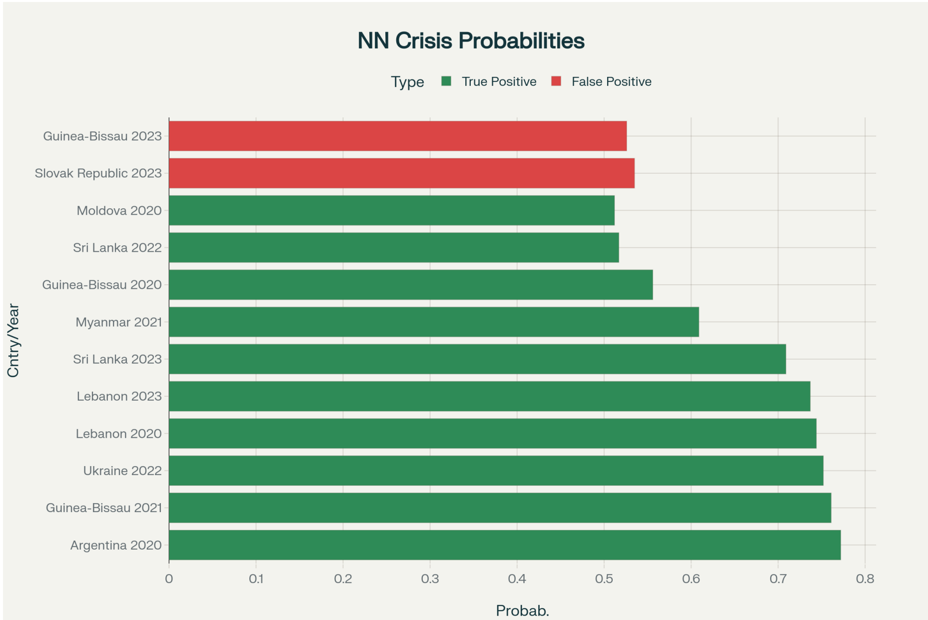

Figure 11: Keras WRN Model Crisis Probabilities.

Figure 12: Keras WRN Model Yearly Crisis Prediction.

8. Limitations

This study advances crisis classification but carries several limitations that should temper interpretation and guide future work.

First, the analysis inherits the Laeven–Valencia crisis dating and country coverage. Their intervention-based rules are systematic but can misdate borderline episodes, truncate long-duration events, or omit policy-light crises, which affects target labeling and downstream performance.

Second, sample and data choices—setting 1996 as the start year and excluding `govt_debt_gdp`—along with country exclusions during reconciliation, constrain external validity. These design decisions may bias results toward better-covered countries and post-1996 dynamics.

Third, the hybrid imputation pipeline (interpolation + Random Forest imputer + median fill) and feature engineering can smooth structural breaks, understate imputation uncertainty, and propagate measurement error into predictors; uncertainty from imputation is not fully propagated into final model inference.

Fourth, modelling choices affect conclusions: SMOTE and synthetic oversampling can distort temporal dependence and inflate apparent validation gains; highly flexible models (WRN) risk overfitting and are less amenable to causal interpretation than the panel logit baseline. Threshold tuning (F_2) improves recall but trades precision in a way sensitive to operational preferences.

Finally, this is a predictive exercise, not a causal identification study: feature importance, PDPs, and SHAP indicate plausibility but do not establish causality. Before operational deployment, we recommend robustness checks (alternative dating rules), formal uncertainty quantification for imputation, temporal-aware resampling, and real-time backtesting.

References

[1] Laeven, L., & Valencia, F. (2020). Systemic Banking Crises Database II. *Journal of Financial Stability*, 45, 100736. https://link.springer.com/article/10.1057/s41308-020-00107-3

[2] Laeven, L., & Valencia, F. (2018). Systemic Banking Crises Revisited. *IMF Working Paper* No. 18/206. International Monetary Fund. https://www.imf.org/en/Publications/WP/Issues/2018/09/14/Systemic-Banking-Crises-Revisited-46232

[3] Laeven, L., & Valencia, F. (2012). Systemic Banking Crises Database: An Update. *IMF Working Paper* No. 12/163. International Monetary Fund. https://www.imf.org/en/Publications/WP/Issues/2016/12/31/Systemic-Banking-Crises-Database-An-Update-26015

[4] Beutel, J., List, S., & von Schweinitz, G. (2019). Does Machine Learning Help Us Predict Banking Crises? *Journal of Financial Intermediation*, 43, 100815. https://www.sciencedirect.com/science/article/abs/pii/S1572308918305801

[5] Peltonen, T. A., Rancan, M., & Sarlin, P. (2017). A New Database for Financial Crises in European Countries. *ECB Occasional Paper* No. 194. European Central Bank. https://www.ecb.europa.eu/pub/pdf/scpops/ecb.op194.en.pdf

[6] Klingebiel, D., & Laeven, L. (2004). Resolving Systemic Financial Crises: Policies and Institutions. *World Bank Policy Research Working Paper* No. 3377. https://openknowledge.worldbank.org/entities/publication/5dc3fba1-1aa5-5115-a44e-17545869b987

[7] Afroz, M. (2025). *Early Warning System for Systemic Banking Crises — Datasets, Analysis Files, and Code*. Available at: https://github.com/22f1000610/EWS